SRM TRP Engineering College
Tiruchirappalli
Course Assignment Instructions with Rubrics

Department	AI & DS							
Programme:	B. Tech	**Semester:**	05		**AY:**	2026 - 2027		
Faculty Handling	Mrs. R. Kalaiyarasi							
Course Code & Title:	AD3501 & Deep Learning							
Assessment Task:	Research Analysis	**Assessment Type:**		Term Work/Self	**Required Hrs:**			
Assessment Question based on:	*Open Ended*	✓	*Problem Based Learning*	✓	*Creative Thinking*			
	Close Ended		*Project Based Learning*		*Critical Thinking*	✓		
	Any Other Type	MCQ or any other	*Challenge Based Learning*	✓	*Level of Thinking (RBT)*	K5		
SDG to be Addressed:	SDG is Required	Yes/No	*SDG Number (Specify or Mention Students Choice)*	4	*Target Number (If Applicable)*			
SDG Integration Level: (If Addressed)	*Level 1: Awareness*		*Level 2: Design Alignment*	✓	*Level 3: Impact and Measurement*			
RIASEC Context	Realistic	Investigative ✓	Artistic	Social	Enterprising	Conventional ✓		
Presentation Format:	PowerPoint/Poster/Video/PDF/Written/Any other (Specify)							
Max Marks:	40							
Relevant Course Outcome(s):	CO1 to CO5							
Submission Deadline:	14.08.2026							

Instructions to the Students:

Assessment Question:

Guidelines:
E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any

Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)

Assignment Question:-

Select a deep learning research topic of your choice and identify one research paper as base paper. Analyse the base paper using an AI tool and document the prompts used. Conduct a literature survey of 10 related research papers, compare their methodologies, datasets, models results and limitations, critically identify and justify a research gap, and map the identified problem to a relevant SDG

Edge device Offline Face recognition Attendance System

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Research Paper Analysis

S. no.	Paper Title	Authors	Journal name	Month&Year	Method & Tech used	Advantages	Limitation
1	DeepFace: Closing the Gap to Human-Level Performance in Face Verification	Yaniv Taigman, Ming Yang, Marc'Aurelio Ranzato, Lior Wolf	CVPR (IEEE)	June 2014	3D face modeling, piecewise affine alignment, 9-layer deep neural network, large-scale supervised training	Established a strong deep-learning face-verification pipeline; explicit alignment improves robustness; demonstrated strong performance on unconstrained faces	Very large model (over 120 million parameters); high training/data requirements; unsuitable for direct deployment on low-resource edge hardware
2	FaceNet: A Unified Embedding for Face Recognition and Clustering	Florian Schroff, Dmitry Kalenichenko, James Philbin	CVPR (IEEE)	June 2015	Deep CNN, direct face embedding, triplet loss, online triplet mining	Produces compact embeddings suitable for recognition, verification and clustering; avoids training a separate classifier for every identity; strong general-purpose representation	Original architecture is not optimized specifically for low-power edge devices; triplet mining and large-scale training are computationally expensive
3	MobileFaceNets: Efficient CNNs for Accurate Real-Time Face Verification on Mobile Devices	Sheng Chen, Yang Liu, Xiang Gao, Zhen Han	Chinese Conference on Biometric Recognition / Springer LNCS	August 2018	Lightweight CNN, bottleneck blocks, depthwise separable operations, ArcFace loss	Designed for mobile and embedded devices; less than 1M parameters; about 4 MB model reported; very low inference latency	Verification performance can degrade with difficult poses, poor image quality and domain shift; lightweight design still requires careful threshold tuning and preprocessing
4	ArcFace: Additive Angular Margin Loss for Deep Face Recognition	Jiankang Deng, Jia Guo, Niannan Xue, Stefanos Zafeiriou	CVPR (IEEE/CVF)	June 2019	Additive Angular Margin Loss, hyperspherical feature learning, deep CNN	Produces highly discriminative and compact identity embeddings; strong intra-class compactness and inter-class separation; negligible computational overhead compared with the backbone	ArcFace is a loss function rather than a complete edge system; final resource use depends on the selected backbone; performance still depends on training data quality and demographic coverage
5	RetinaFace: Single-stage Dense Face Localisation in the Wild	Jiankang Deng, Jia Guo, Yuxiang Zhou, Jinke Yu, Irene Kotsia, Stefanos Zafeiriou	CVPR (IEEE/CVF)	June 2020	Single-stage face detector, multi-task learning, five facial landmarks, lightweight backbones	Strong face detection across scales and challenging conditions; provides landmarks useful for alignment; lightweight	Detection remains sensitive to very small, blurred or heavily occluded faces; landmark/detection stages add computation before recognition; complete pipeline must still be

						backbones can run in real time on CPU	optimized for edge hardware
6	AdaFace: Quality Adaptive Margin for Face Recognition	Minchul Kim, Anil K. Jain, Xiaoming Liu	CVPR (IEEE/CVF)	June 2022	Quality-adaptive margin loss, feature-norm-based image quality estimation	Explicitly accounts for image quality during training; improves recognition on low-quality and difficult face datasets; useful for cameras with variable image quality	Introduces training-loss complexity rather than solving the complete low-power inference problem; requires suitable training data and quality variation to obtain the full benefit
7	EdgeFace: Efficient Face Recognition Model for Edge Devices	Anjith George, Christophe Ecabert, Hatef Otroshi Shahreza, Ketan Kotwal, Sebastien Marcel	IEEE Transactions on Biometrics, Behavior, and Identity Science	April 2024	EdgeNeXt-inspired CNN–Transformer hybrid, low-rank linear layer, compact embedding network	Specifically designed for edge devices; 1.77M parameters; reported 99.73% LFW accuracy and strong IJB-B/IJB-C results; good balance between accuracy, model size and computation	Benchmark performance does not guarantee reliability in rural school conditions; real-world illumination, camera quality, child-age variation and open-set identity handling need additional validation
8	Lightweight Face Recognition-Based Portable Attendance System with Liveness Detection	Nico Surantha, Boy Sugijakko	Internet of Things (Elsevier)	April 2024	Raspberry Pi, MobileNetV2, transfer learning, CNN-based liveness detection, face recognition attendance	Directly relevant to portable attendance; designed for limited computational power; reported average processing below 0.6 s and 96% live-subject accuracy; addresses spoofing	Spoof-detection accuracy decreased for harder attack levels; added liveness detection increases processing cost; Raspberry Pi results may not transfer directly to every edge device or school environment
9	Enhancing Classroom Attendance Systems with Face Recognition through CCTV using Deep Learning	Authors as reported by the publisher	Procedia Computer Science (Elsevier)	2025	CCTV classroom images, data augmentation, transfer learning, ResNet-50, VGG-16, PCA	Demonstrates practical classroom attendance using deep face recognition; reported strong precision/recall for ResNet-50; evaluates data from actual classroom environments	Depends on camera quality and viewing geometry; heavier models are less suitable for low-power offline deployment; classroom-specific evaluation may not generalize to rural schools
10	LaED: A Novel Lightweight, Edge-Aware and Explainable Deep Learning Model for Privacy-Preserving Facial Attendance Tracking in Resource-Constrained Educational Environments	Esther Omolara Abiodun, Oludare Isaac Abiodun, Moatsum Alawida, Bayan Abu Shawar, Abid Mehmood et al.	Scientific Reports (Nature)	May 2026	Lightweight edge AI, multimodal spoof detection, open-set recognition, fairness-aware learning, federated learning, differential privacy, explainability	Combines edge inference, spoof resistance, unknown-identity rejection, privacy protection and fairness; reports sub-150 ms inference on edge hardware	More complex than a minimal offline attendance device; federated learning and privacy mechanisms add engineering overhead; reported results require validation on the target school's devices, connectivity pattern and

							demographic distribution

Research Gap

1. Real-World Environmental Robustness

Most face recognition systems report strong results on controlled or benchmark datasets, but practical school deployment introduces illumination changes, low-light images, motion blur, camera quality variation, different face-to-camera distances, partial occlusion, head-pose changes, and crowded scenes. The reviewed literature shows that detection, alignment, image quality and embedding quality remain tightly coupled. A rural-school system therefore needs explicit quality checks and robust preprocessing rather than relying only on benchmark accuracy.

2. Computational and Energy Constraints on Edge Devices

Large CNN and hybrid models can provide high recognition accuracy but may require substantial memory, compute and power. MobileFaceNets and EdgeFace demonstrate that compact models can retain strong recognition performance, while portable attendance research shows that lightweight models can operate on Raspberry Pi-class hardware. However, there is still a practical gap between an efficient model and a complete low-power attendance pipeline that includes detection, alignment, liveness, matching, storage and user interface.

3. Offline-First Operation and Intermittent Connectivity

Most published attendance systems assume continuous connectivity, centralized servers, or cloud-backed processing. This is a poor fit for schools where internet access may be intermittent or unavailable. The literature supports edge inference and local processing, but a complete offline-first architecture with local identity templates, attendance records, secure storage and optional later synchronization requires further system-level validation.

4. Child-Focused and Demographically Diverse Recognition

Many face-recognition benchmarks contain adult-centric or broad internet datasets rather than school-specific populations. Student faces also change with age, hairstyle, accessories and growth. Recognition thresholds and enrollment strategies that work on general benchmarks may therefore not transfer directly to rural school populations. The target population should be evaluated separately for age, gender, skin tone, camera conditions and intra-person variation.

5. Spoofing, Proxy Attendance and Unknown-Identity Rejection

A face matcher alone can be vulnerable to printed photographs, phone-screen replays or other presentation attacks. The portable attendance literature demonstrates the value of liveness detection, while recent educational research adds open-set recognition to reject

identities that are not enrolled. A rural-school system needs an appropriate balance between spoof resistance, latency, cost and ease of use.

6. Privacy and Local Biometric Data Protection

Face recognition processes biometric information that requires careful handling. Cloud transmission increases exposure and creates dependence on connectivity. Edge-only processing can reduce unnecessary transmission, but local storage must still be protected through encryption, access control, secure deletion, minimal retention and clear consent/notification procedures. Privacy should therefore be treated as an architectural requirement rather than an optional feature.

7. Robustness, Fairness and Explainability

Recent research increasingly considers fairness, explainability and privacy, but these properties are not consistently evaluated together with edge efficiency. A school deployment should measure false accept and false reject behavior across relevant demographic and environmental groups, provide understandable failure reasons such as poor lighting or insufficient face quality, and retain auditable attendance events without storing unnecessary images.

Identified Research Gap

Based on the literature survey, the key research gap is not simply higher face-recognition accuracy. It is the development of a complete offline, lightweight, privacy-aware and spoof-resistant face-recognition attendance system that can operate reliably on low-cost edge hardware in rural schools. Existing work separately addresses efficient recognition, liveness detection, classroom attendance, privacy, fairness and edge AI. A unified system must combine these capabilities while handling intermittent connectivity, limited power and storage, variable camera quality, child-specific facial variation, unknown identities and secure local attendance storage.

Proposed Future Research Direction

Develop and benchmark a lightweight face-recognition pipeline using an efficient detector, alignment module and compact embedding model such as EdgeFace or MobileFaceNet.
Use quality-aware recognition and adaptive thresholds to handle low-light, blur, pose and low-resolution classroom images.
Add lightweight liveness detection to reduce proxy attendance and presentation attacks without requiring expensive additional sensors.
Design an offline-first local database for encrypted face embeddings and attendance records, with optional synchronization only when connectivity becomes available.
Evaluate open-set recognition so that unknown or unregistered students are rejected instead of being forced into the nearest enrolled identity.

Create a school-specific evaluation dataset with consent and measure accuracy, false acceptance, false rejection, latency, memory use, energy use and robustness across realistic classroom conditions.
Investigate privacy-preserving and fairness-aware approaches, including local-only processing, minimal biometric retention, secure deletion and demographic performance auditing.

Sustainable Development Goal (SDG) Mapping

The selected research topic, Offline Edge-AI Face Recognition for Rural School Attendance, supports the United Nations Sustainable Development Goals through improved educational administration, accessible digital infrastructure and responsible AI deployment. The mapping below is focused on the actual research problem rather than generic applications of facial recognition.

Primary SDG: SDG 4 – Quality Education

Why SDG 4?

Reliable attendance records help schools monitor student participation and identify attendance patterns without requiring teachers to spend substantial classroom time on manual roll calls. An offline system is particularly relevant where internet connectivity is unreliable.

Contribution

Automates routine attendance marking and reduces manual recording effort.
Supports timely attendance records for teachers and school administrators.
Reduces opportunities for proxy attendance when liveness and identity verification are used.
Continues operating during internet outages because recognition and attendance storage occur locally.
Provides a foundation for future analytics on attendance and student participation without requiring continuous cloud connectivity.

Justification: The reviewed attendance studies demonstrate that face recognition can automate classroom attendance, while edge-oriented research shows that recognition can be moved onto resource-constrained devices.

Secondary SDG: SDG 9 – Industry, Innovation and Infrastructure

Why SDG 9?

The project applies compact deep-learning models, edge computing, local databases and efficient inference to create useful AI infrastructure under resource constraints. This directly addresses the practical challenge of deploying AI where cloud infrastructure and reliable connectivity are limited.

Contribution

Develops lightweight AI suitable for low-cost edge hardware.
Reduces dependence on continuous cloud inference and network availability.
Encourages efficient computing through compact models and local processing.
Creates an extensible edge-AI architecture for other rural digital services.
Promotes responsible deployment of computer vision in resource-constrained environments.

Justification: EdgeFace, MobileFaceNets and recent educational attendance research show the growing importance of efficient, deployable face-recognition systems rather than accuracy-only models.

Additional Relevant SDGs

SDG 10 – Reduced Inequalities

A low-cost offline system can reduce the infrastructure gap between schools with reliable high-speed connectivity and schools operating with limited network access. Fairness evaluation is also important so that recognition quality does not systematically disadvantage particular student groups.

SDG 16 – Peace, Justice and Strong Institutions

Secure, auditable and tamper-resistant attendance records can support transparent school administration. Privacy-preserving design, access control and clear accountability are essential because the system handles sensitive biometric information.

References (IEEE Format)

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[6] M. Kim, A. K. Jain, and X. Liu, “AdaFace: Quality Adaptive Margin for Face Recognition,” in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2022, pp. 18750–18759. doi: 10.1109/CVPR52688.2022.01820.

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[9] “Enhancing Classroom Attendance Systems with Face Recognition through CCTV using Deep Learning,” Procedia Computer Science, vol. 258, pp. 3031–3041, 2025. doi: 10.1016/j.procs.2025.04.561.

[10] E. O. Abiodun, O. I. Abiodun, M. Alawida, B. Abu Shawar, and A. Mehmood et al., “LaED: A Novel Lightweight, Edge-Aware and Explainable Deep Learning Model for Privacy-Preserving Facial Attendance Tracking in Resource-Constrained Educational Environments,” Scientific Reports, vol. 16, Art. no. 15215, May 2026. doi: 10.1038/s41598-026-42051-8.